

Patient A 500-Year Hardening Plan (Long-Form Vitals Edition)

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Classification: Alias-only, privacy-preserving

Subject: Patient A

Founder Abstract

This document defines a 500-year hardening architecture for Patient A that treats safety, evidence, and continuity as a single system. The central objective is to keep care quality stable across tool changes, staff turnover, governance shifts, and environmental disruption.

Vitals-First Mission

- Preserve life-critical continuity of care.
- Keep all high-impact decisions traceable to evidence.
- Prevent drift from safety doctrine across generations.
- Sustain financial and operational reliability without identity leakage.

Core Vitals Framework

Physiologic Vitals

- Cardiovascular stability envelope.
- Respiratory resilience envelope.
- Neurologic baseline consistency profile.
- Metabolic and renal trend integrity.
- Inflammation and immune variance tracking.

Safety Vitals

- Contraindication detection sensitivity.
- Drug-interaction and protocol-collision detection.
- Escalation latency for red-flag events.
- False reassurance suppression index.

System Vitals

- Evidence freshness half-life.
- Control drift index.
- Incident recurrence rate.
- Recovery time to stable baseline.
- Succession readiness score.

Trust Vitals

- Provenance completeness rate.
- Audit replay success rate.
- Consent continuity confidence.
- Oversight quorum reliability.

12-Pronged Hardening Architecture

1. Identity and consent integrity.
 - Alias-only records; no direct identifiers in operating artifacts.
 - Multi-party consent checkpoints for irreversible actions.
 - Rotating consent attestation with immutable decision log.
2. Evidence provenance chain.
 - Every critical assertion linked to source and review date.
 - Evidence quality tiers and stale-evidence timeouts.
 - Automatic downgrade of confidence when provenance weakens.

3. Clinical risk gates.

- Fail-closed rule under uncertain contraindications.
- Mandatory second-pass review on red-zone vitals.
- Decision freeze when conflicting evidence exceeds threshold.

4. Data minimization boundaries.

- Least-privilege access by role and context.
- Split storage domains for sensitive and operational data.
- Periodic purge/retention compliance checks.

5. Model and tool trust boundaries.

- Capability allowlists for each agent/tool role.
- Human sign-off before high-impact protocol changes.
- Mismatch alarms when tool output exceeds confidence band.

6. Change control and release governance.

- Two-person approval for safety-impacting updates.
- Pre-release checklist mapped to hardening policy.
- Rollback-ready release packages for all critical updates.

7. Incident response and recovery.

- Triage tiers with explicit owner-on-call mapping.
- Evidence snapshot at incident start and closure.
- Post-incident correction loop with recurrence controls.

8. Financial continuity controls.

- Minimum runway and reserve covenant thresholds.
- Donation/funding intake controls with reconciliation checks.
- Risk limits for single-source funding dependence.

9. Legal/regulatory adaptability.

- Jurisdictional policy overlays and periodic legal review.
- Evidence packet format ready for external audits.
- Escalation path for regulatory conflict.

10. Human oversight and succession.

- Operator competency matrix and renewal cadence.
- Shadow-to-lead transition playbooks.
- Succession simulation every five years.

11. Infrastructure resilience.

- Backup/restore verification cadence.
- Offline-safe operating fallback for critical workflows.
- Integrity checksums for long-memory archives.

12. Anti-drift long-memory doctrine.

- Canonical doctrine snapshots with signed attestations.
- Drift-detection diffs across policy and implementation.
- Governance renewal without provenance loss.

Epoch Roadmap

Epoch 0: Years 0-2 (Stabilization)

Objectives:

- Establish baseline for all physiologic, safety, system, and trust vitals.
- Implement minimum viable controls for each prong.
- Validate fail-closed gate behavior with simulation drills.

Entry criteria:

- Alias policy active.
- Baseline vitals schema approved.
- Incident ownership roster assigned.

Exit criteria:

- 95% provenance completeness for high-impact decisions.
- Red-zone escalation under target latency.
- Quarterly gate pass rate above threshold.

Epoch 1: Years 3-10 (Institutional Hardening)

Objectives:

- Convert tribal knowledge into enforceable controls.
- Add independent review and conflict-of-interest checks.
- Build annual doctrine stress test discipline.

Entry criteria:

- Epoch 0 closure attestations complete.
- Multi-role review workflow active.

Exit criteria:

- Sustained low drift index.
- Repeatable audit replay success.
- Incident recurrence below set ceiling.

Epoch 2: Years 10-50 (Intergenerational Continuity)

Objectives:

- Preserve policy meaning across staff and tool transitions.
- Formalize compatibility and migration standards.
- Keep evidence chains intact through platform shifts.

Entry criteria:

- Succession program operational.
- Long-memory archive standard in production.

Exit criteria:

- Cross-generation handoff pass rate stable.
- Historical evidence replay integrity maintained.

Epoch 3: Years 50-500 (Civilizational Durability)

Objectives:

- Maintain low-assumption doctrine readability.
- Re-validate axioms against emerging risks.
- Sustain governance with minimal identity exposure.

Entry criteria:

- Doctrine snapshots and checksum lineage complete.
- Legal and ethical review cadence institutionalized.

Exit criteria:

- Continuous continuity with no unresolved critical drift events.

Gate and Scoring Model

Each gate includes:

- Gate ID
- Objective

- Required evidence bundle
- KPI thresholds
- Fail conditions
- Escalation owner
- Recovery protocol reference

Scoring bands:

- Green: Safe to proceed.
- Yellow: Proceed with mitigation and review.
- Red: Stop; fail closed until corrected.

Vitals Measurement Cadence

- Real-time: high-risk safety signals.
- Daily: physiologic trend summary and anomaly scan.
- Weekly: system/trust vitals rollup.
- Monthly: gate readiness and drift scan.
- Quarterly: convergence audit.
- Annual: full doctrine stress test.

Incident Playbook Skeleton

Phase 1 Detection:

- Trigger source, severity, and confidence capture.

Phase 2 Containment:

- Freeze high-risk actions.
- Activate fallback pathways.

Phase 3 Validation:

- Confirm evidence integrity and causal chain.

Phase 4 Recovery:

- Restore baseline with monitored checkpoints.

Phase 5 Learning:

- Record root cause and policy correction diff.

Financial Hardening Layer

- Maintain operating reserve thresholds.
- Maintain donor transparency logs with reconciliation checks.
- Require dual-review for major disbursement decisions.
- Track concentration risk and diversification targets.

Governance and Roles

- Founder/Steward: doctrine guardian and tie-break authority.
- Clinical Safety Lead: red-zone escalation owner.
- Evidence Lead: provenance quality owner.
- Operations Lead: continuity and incident commander.
- Audit Lead: independent verification owner.

Documentation Pack Requirements

- Canonical plan source.
- Quarterly convergence report.
- Annual attestation report.
- Incident ledger and corrective-action registry.
- Succession readiness packet.

Acceptance Criteria

- All 12 prongs mapped to controls, KPIs, and failure modes.
- All 4 epochs include entry/exit criteria.
- All vitals classes have cadence and owners.
- Alias-only identity policy maintained end-to-end.
- PDF output matches canonical source ordering.

Closing Principle

Truth before speed. Safety before narrative. Evidence before confidence. Continuity before convenience.